

The Characteristic Equation

Geometric Algorithms

Practice Problem

$$\begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

*Determine the dimension of the eigenspace of A
for the eigenvalue 4* $\text{Nul}(A - 4I)$

(try not to do any row reductions)

Answer

$$\text{Nul}(A - 4I)$$

$$(A - 4I)\vec{v} = \vec{0}$$

$$A\vec{v} - 4\vec{v} = \vec{0}$$

$$A\vec{v} = 4\vec{v}$$

$$A - 4I = \begin{bmatrix} 1 & 2 & 3 & 0 \\ -1 & -2 & -3 & 1 \\ 2 & 4 & 6 & 0 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$\vec{v}_1$ points to the first column, $\vec{v}_4$ points to the fourth column.

$$\begin{bmatrix} 5 & 2 & 3 & 0 \\ -1 & 2 & -3 & 1 \\ 2 & 4 & 10 & 0 \\ 1 & 2 & 3 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{rank}(A - 4I) &= \# \text{ pivots} \\ &= \dim(\text{Col}(A - 4I)) \\ &= 2 \end{aligned}$$

$$\text{rank}(A) + \dim(\text{Nul}(A)) = n \text{ (\# cols)}$$

$$\dim(\text{Nul}(A - 4I)) = 4 - 2 = 2$$

Objectives

- » Briefly recap eigenvalues and eigenvectors
- » Get a primer on **determinants**
- » Determine how to **find eigenvalues** (not just verify them)

Keyword

eigenvectors

eigenvalues

eigenspaces

eigenbases

determinant

characteristic equation

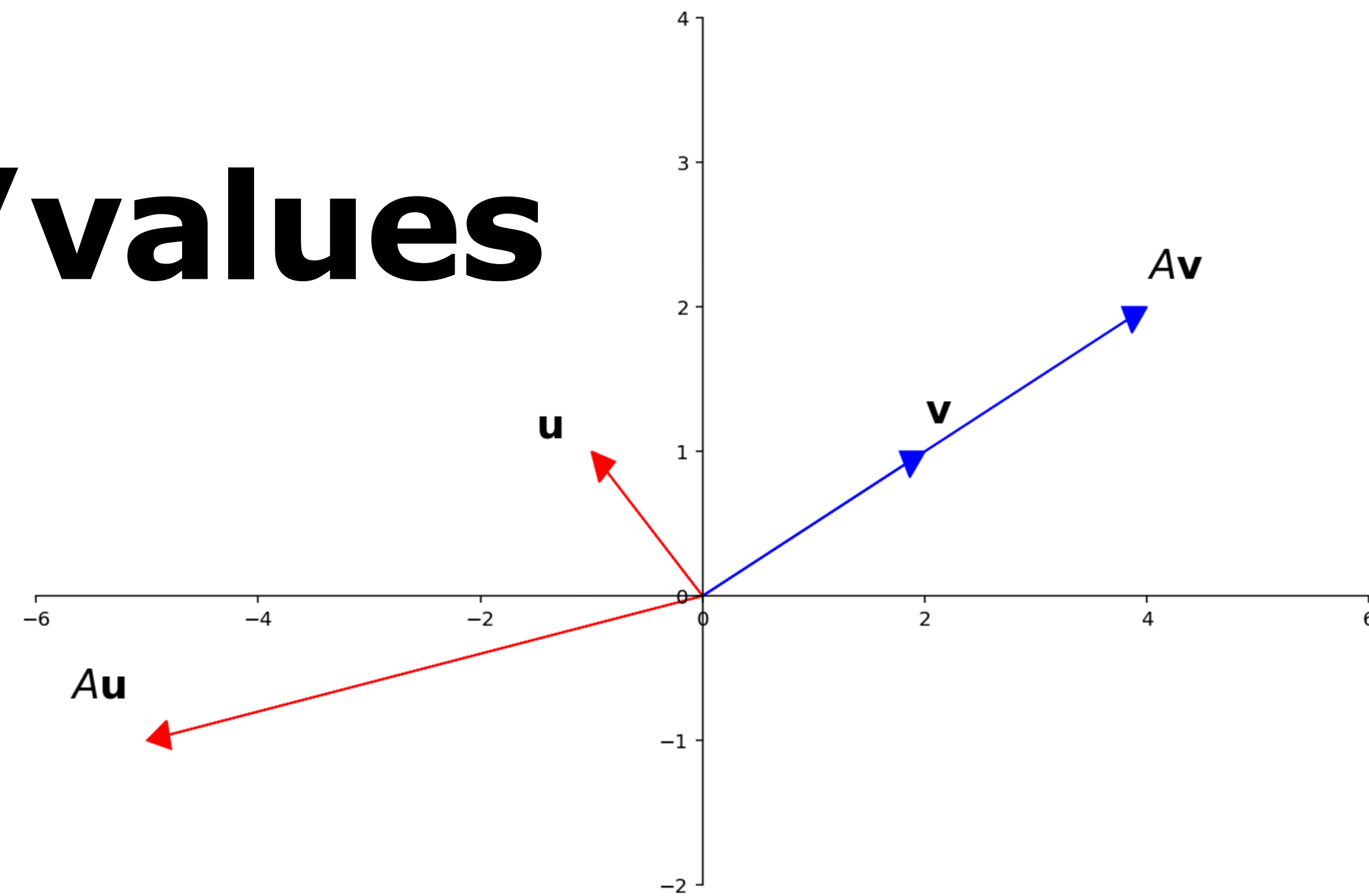
polynomial roots

triangular matrices

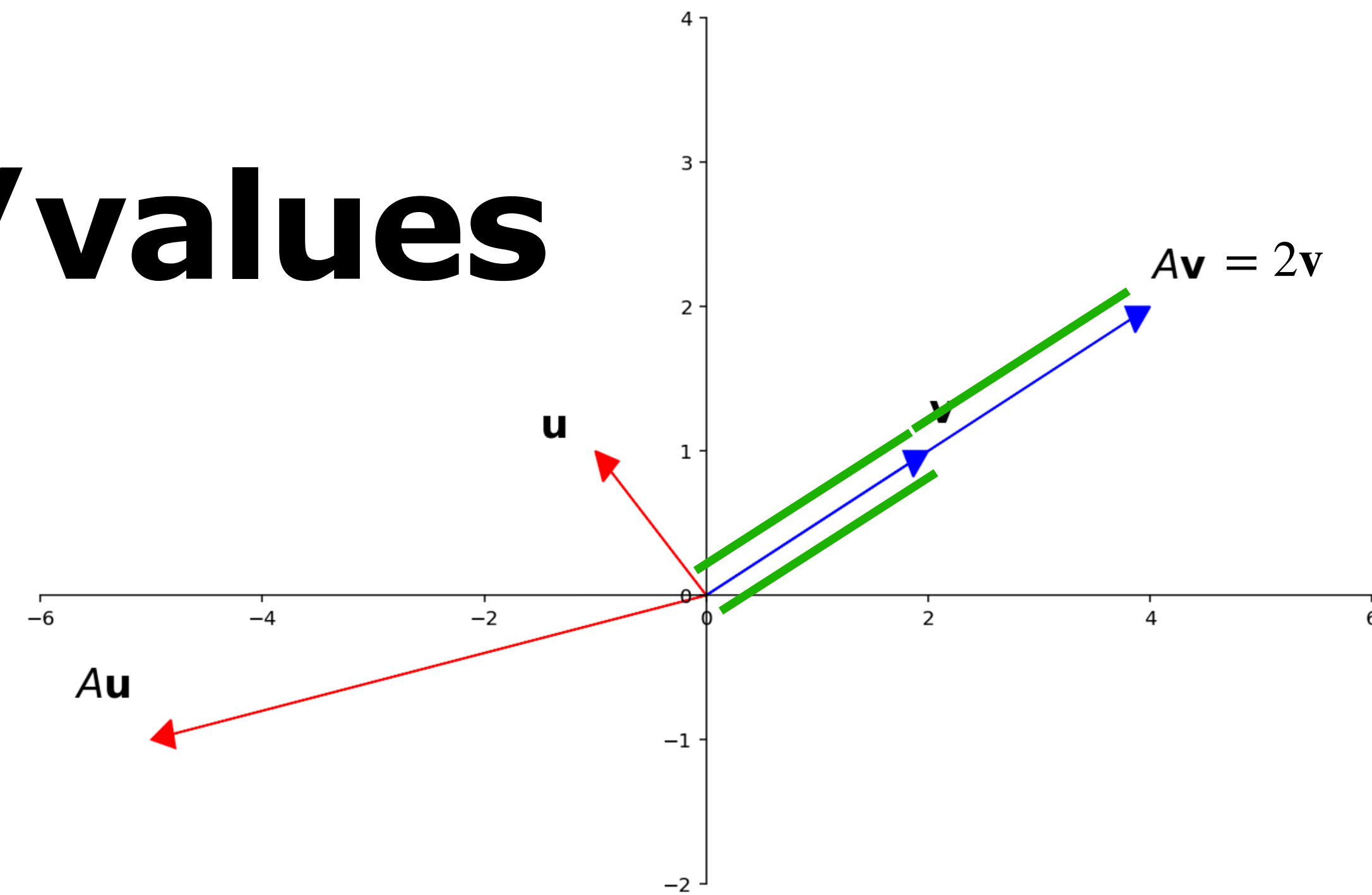
multiplicity

Recap

Recall: Eigenvectors/values



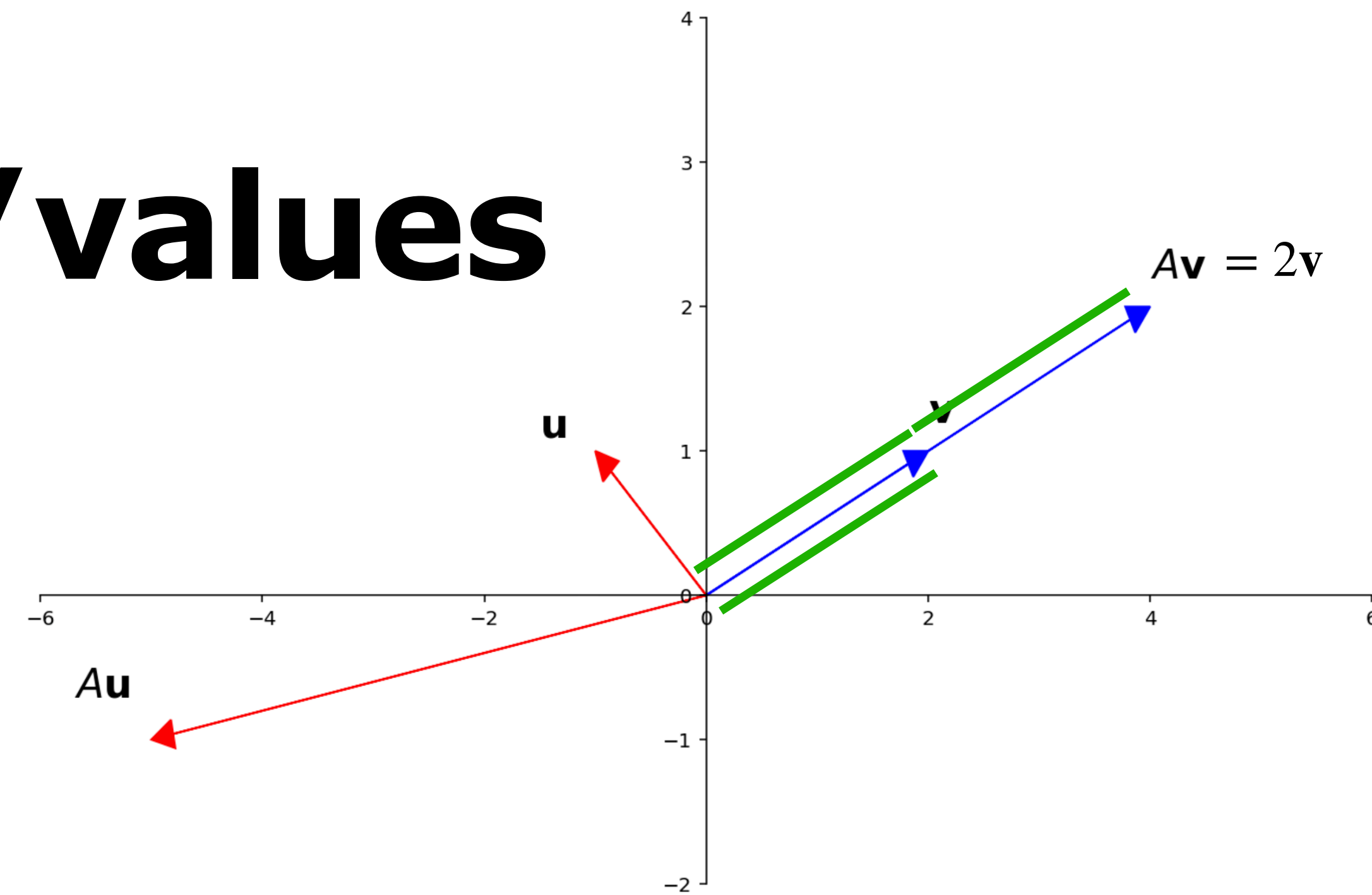
Recall: Eigenvectors/values



Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector** and **eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if

$$A\mathbf{v} = \lambda\mathbf{v}$$

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Eigenvectors are just scaled by A

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Ex.

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} \quad \checkmark$$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 9 \end{bmatrix} \quad \times$$

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*If we don't need the vector we can just show that $A - \lambda I$ is **not** invertible (by IMT)*

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(we did this for our recap problem)

Recall: Eigenvectors and Linear Dynamical Systems

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$ of A with eigenvalues $\lambda_1 > \lambda_2 \dots \geq \lambda_k$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

$$\begin{aligned} \vec{v}_k &= A^k \vec{v}_0 \\ &= A^k (\alpha_1 \vec{b}_1 + \dots + \alpha_n \vec{b}_n) \\ &= \alpha_1 A^k \vec{b}_1 + \dots + \alpha_n A^k \vec{b}_n \\ &= \alpha_1 \lambda_1^k \vec{b}_1 + \dots + \alpha_n \lambda_n^k \vec{b}_n \end{aligned}$$

$$\lim_{k \rightarrow \infty} \frac{v_k}{\lambda_1^k c_1 \vec{b}_1} = 1$$

$$\frac{\alpha_1 \lambda_1^k \vec{b}_1 + \dots + \alpha_n \lambda_n^k \vec{b}_n}{\alpha_1 \lambda_1^k \vec{b}_1} = 1 + \sum_{i=2}^n \frac{\alpha_i}{\alpha_1} \left(\frac{\lambda_i}{\lambda_1} \right)^k \vec{b}_i$$

Eigenbases

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*We can represent vectors as **unique** linear combinations of eigenvectors*

Not all matrices have eigenbases

Eigenbases and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if A has an eigenbasis $\mathbf{b}_1, \dots, \mathbf{b}_k$, then

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for some constant c_1 , where λ_1 is the **largest eigenvalue** of A and $\mathbf{b}_1$ is its **eigenvector**

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for some constant c_1 , where λ_1 is the **largest eigenvalue** of A and $\mathbf{b}_1$ is its **eigenvector**

The largest eigenvalue describes the long-term exponential behavior of the system

Finding Eigenvalues

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A. Can we somehow "solve for λ " in the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

Determinants

An Aside: Determinants are Mysterious

Determinants are strangely polarizing

Some people love them, some people hate them

We'll only scratch the surface...

Down with Determinants!

Sheldon Axler



102 (1995), 139-154.

ry writing from the Mathematical Association of America.

without determinants. The standard proof that a square matrix of complex numbers has an eigenvalue uses d
erminants, this allows us to define the multiplicity of an eigenvalue and to prove that the number of eigenva
haracteristic and minimal polynomials and then prove that they behave as expected. This leads to an easy p
determinants, this paper gives a simple proof of the finite-dimensional spectral theorem.

n this paper. The book is intended to be a text for a second course in linear algebra.

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A determinant, written $\det(A)$, is a *number* associated with a matrix

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In broad strokes, it's a big sum of products of entries of A

A Scary-Looking Definition (we won't use)

$$\det(A) = \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)}$$

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We can think of this as a procedure

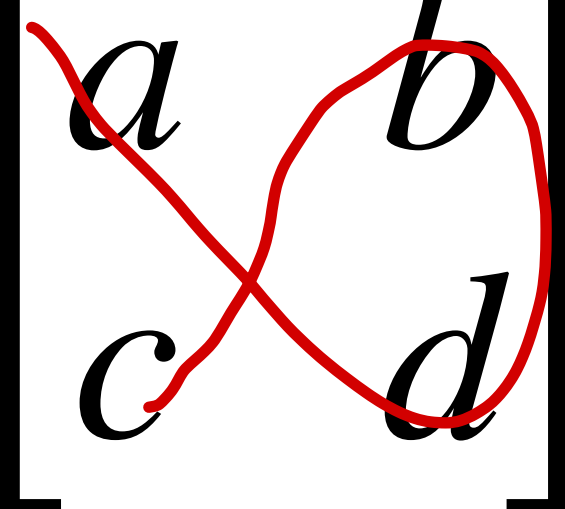
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```
1 FUNCTION det(A):  
2   total = 0  
3   FOR all matrix B we can get by swapping a bunch of rows of A:  
4     s = 1 IF (# of swaps necessary) is even ELSE -1  
5     total += s * (product of the diagonal entries of B)  
6   RETURN total
```

The Determinant of 2×2 Matrices

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$


$(-1)^0 ad$ is the product of the diagonal

$(-1)^1 bc$ is the product of the diagonal after 1 swap

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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow^0 \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

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$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

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$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow^2 \begin{bmatrix} g & h & i \\ a & b & c \\ d & e & f \end{bmatrix}$$

$$(-1)^2 gbf$$

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$$(-1)^1 ahf$$

Another Perspective

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{R_2 \leftarrow aR_2} \begin{bmatrix} a & b \\ ac & ad \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - cR_1}$$

$$\sim \begin{bmatrix} a & b \\ 0 & ad - bc \end{bmatrix}$$

$$ad - bc = 0 \iff A \text{ is invertible}$$

Another Perspective

$$\begin{bmatrix} a & b & c \\ ad & ae & af \\ ag & ah & ai \end{bmatrix} \sim \begin{bmatrix} a & b & c \\ 0 & ae - db & af - dc \\ 0 & ah - gb & ai - gc \end{bmatrix} \rightarrow A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

B

$$\begin{bmatrix} a & b & c \\ 0 & ae - db & af - dc \\ 0 & 0 & \det(B) \end{bmatrix}$$

ex.

$$\det(B) = \det(A)$$

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So we can extend the IMT:

» A is invertible

» $\det(A) \neq 0$

(these must be all true or all false)

» 0 is not an eigenvalue

Determinants (the definition we'll use)

$$\det(A) = \frac{(-1)^s}{c} U_{11} U_{22} \cdots U_{nn}$$

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0 if A is not invertible

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Example

$$S = \cancel{0} 1$$

$$C = 1$$

$$\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & -6 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$



$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & -1 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 3R_2$$



$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\det(A) = \frac{(-1)^1}{1} (1) (-2) (-1) = -2$$

Example (Again)

$$S = \cancel{0} \cancel{1} 2$$

$$c = \cancel{1}$$

$$\begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

$$R_1 \leftrightarrow R_2 \rightarrow \begin{bmatrix} 2 & 4 & -1 \\ 1 & 5 & 0 \\ 0 & -2 & 0 \end{bmatrix}$$

$$\frac{(-1)^2}{\left(-\frac{1}{4}\right)} (1)(1)\left(\frac{1}{2}\right)$$

$$\frac{\frac{1}{2}}{-1/4} = -2 \quad c = \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) = -\frac{1}{4}$$

$$R_1 \leftarrow \boxed{\frac{1}{2}} R_2 \rightarrow \begin{bmatrix} 1 & 2 & -1/2 \\ 0 & -2 & 0 \\ 0 & 3 & 1/2 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - R_1 \rightarrow \begin{bmatrix} 1 & 1 & -1/2 \\ 0 & 3 & 1/2 \\ 0 & -2 & 0 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3 \rightarrow \begin{bmatrix} 1 & 2 & -1/2 \\ 0 & -2 & 0 \\ 0 & 3 & 1/2 \end{bmatrix}$$

$$R_2 \leftarrow \boxed{-\frac{1}{2}} R_2 \rightarrow \begin{bmatrix} 1 & 2 & -1/2 \\ 0 & 1 & 0 \\ 0 & 3 & 1/2 \end{bmatrix}$$

$$R_3 \leftarrow R_3 - 3R_2 \rightarrow \begin{bmatrix} \boxed{1} & 2 & -1/2 \\ 0 & \boxed{1} & 0 \\ 0 & 0 & \boxed{1/2} \end{bmatrix}$$

The definition holds no matter
which row operations you use

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2. Keep track of the number of row swaps you used, call this s , and the product of all scalings, call this c
3. Determine the product of entries along the diagonal of U , call this P
4. The determinant of A is $\frac{(-1)^s P}{c}$

The Shorter Version

`numpy.linalg.det(A)`

Beyond small matrices, we'll just use a computer

Properties of Determinants

Properties of Determinants (1)


$$\det(AB) = \det(A) \det(B)$$

Cor. AB is invertible $\Leftrightarrow$ if A and B are invertible

Question

Use the fact that $\det(AB) = \det(A)\det(B)$ to give an expression for $\det(A^{-1})$ in terms of $\det(A)$

Hint. What is $\det(I)$?

$$\det(A^{-1}A) = \det(I) = 1$$

||

$$\det(A^{-1})\det(A)$$
$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Properties of Determinants (2)

$$\det(A^T) = \det(A)$$

It follows that A^T is invertible if and only if A is invertible

(we also won't verify this)

Question

If $A^{-1} = A^T$, then what is $\det(A)$?

$$\frac{1}{\det(A)} = \det(A^{-1}) = \det(A^T) = \det(A)$$

$$(\det(A))^2 = 1$$

$$\det(A) = 1$$

or

$$\det(A) = -1$$

Properties of Determinants (3)

Thm. If A is triangular, then $\det(A)$ is the product of entries along the diagonal

$$\det \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix} = a_{11} a_{22} \cdots a_{nn} = \det \begin{bmatrix} a_{11} & & 0 \\ & a_{22} & \\ \vdots & & \ddots \\ 0 & & & a_{nn} \end{bmatrix}$$

Question

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

Find the determinant of the above matrix

Answer

$$\begin{bmatrix} 1 & 5 & -4 \\ -1 & -5 & 5 \\ -2 & -8 & 7 \end{bmatrix}$$

Characteristic Equation

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But a matrix may not have numbers as entries

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We might think of the matrix $A - \lambda I$ as having *polynomials* as entries

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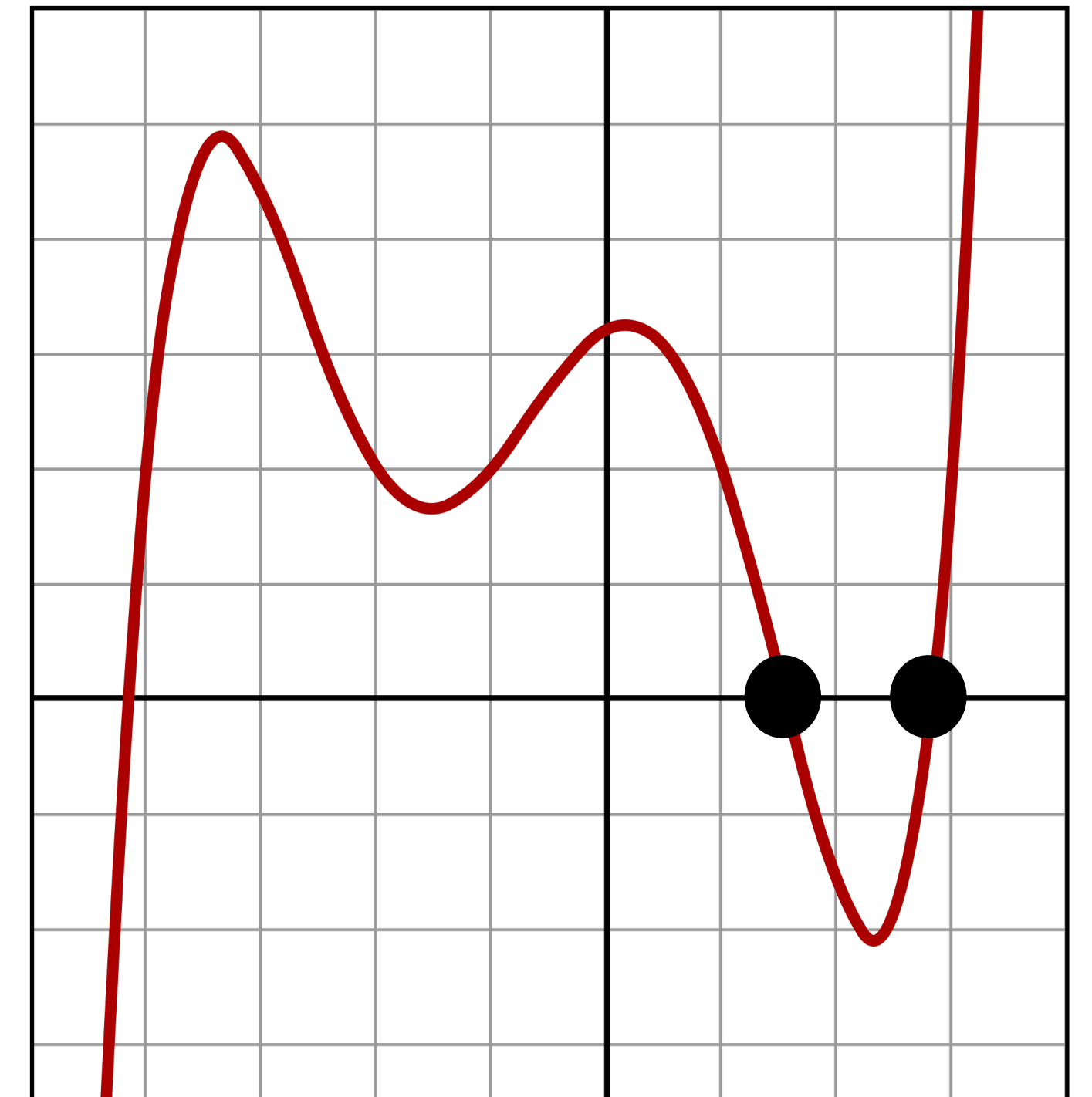
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But a matrix may not have numbers as entries

We might think of the matrix $A - \lambda I$ as having *polynomials* as entries

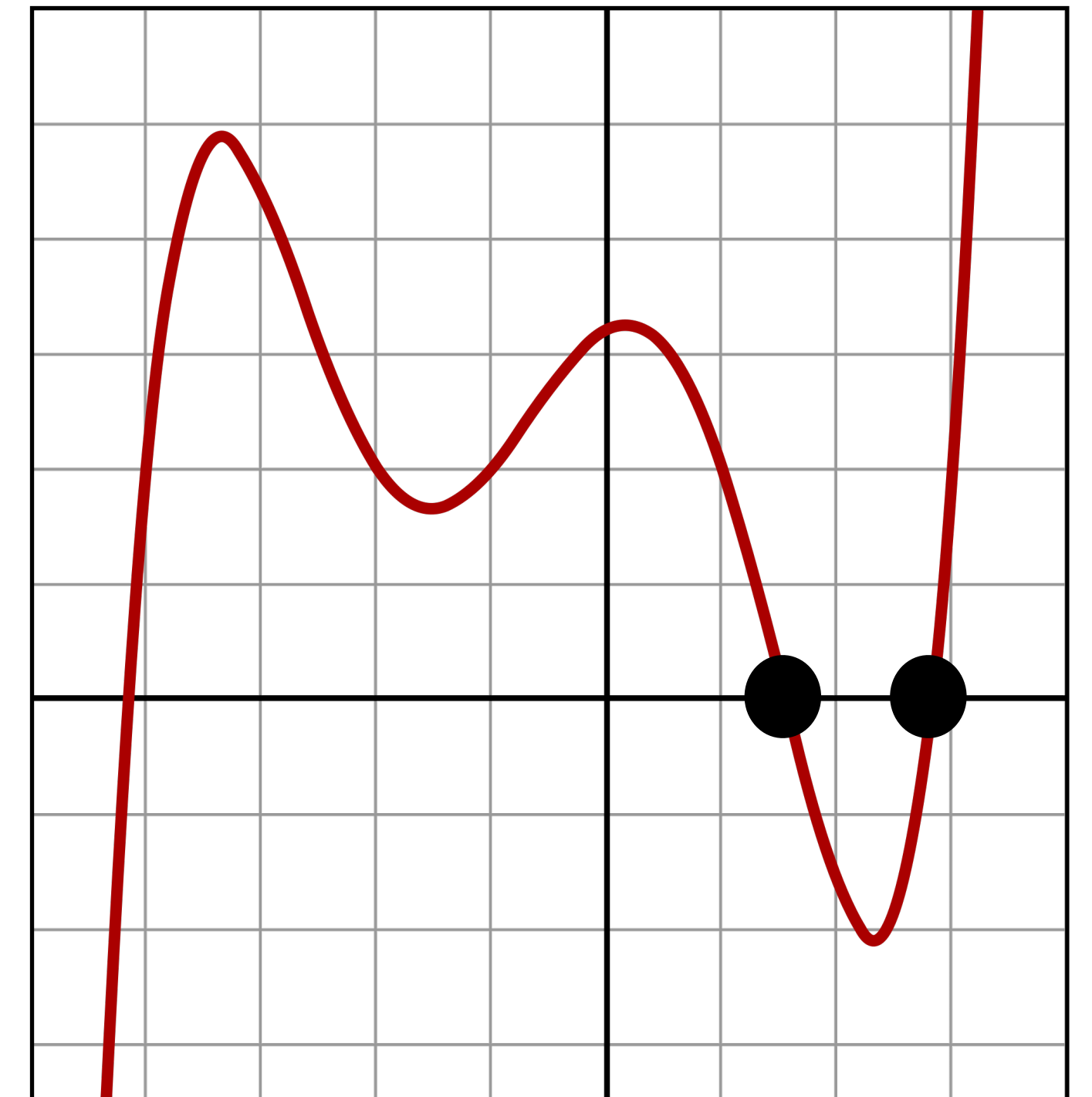
Then $\det(A - \lambda I)$ is a **polynomial**

Reminder: Polynomial Roots



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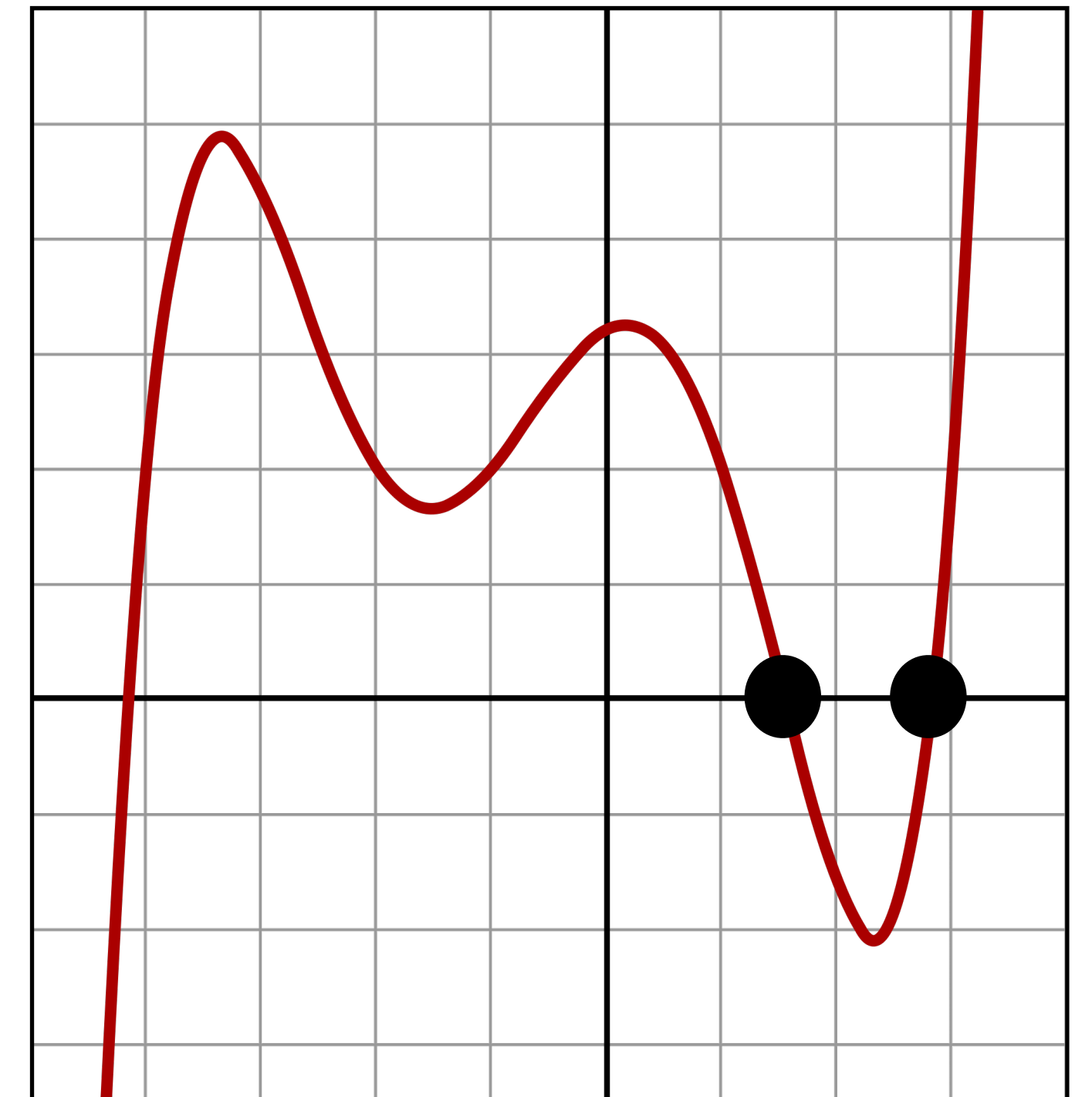
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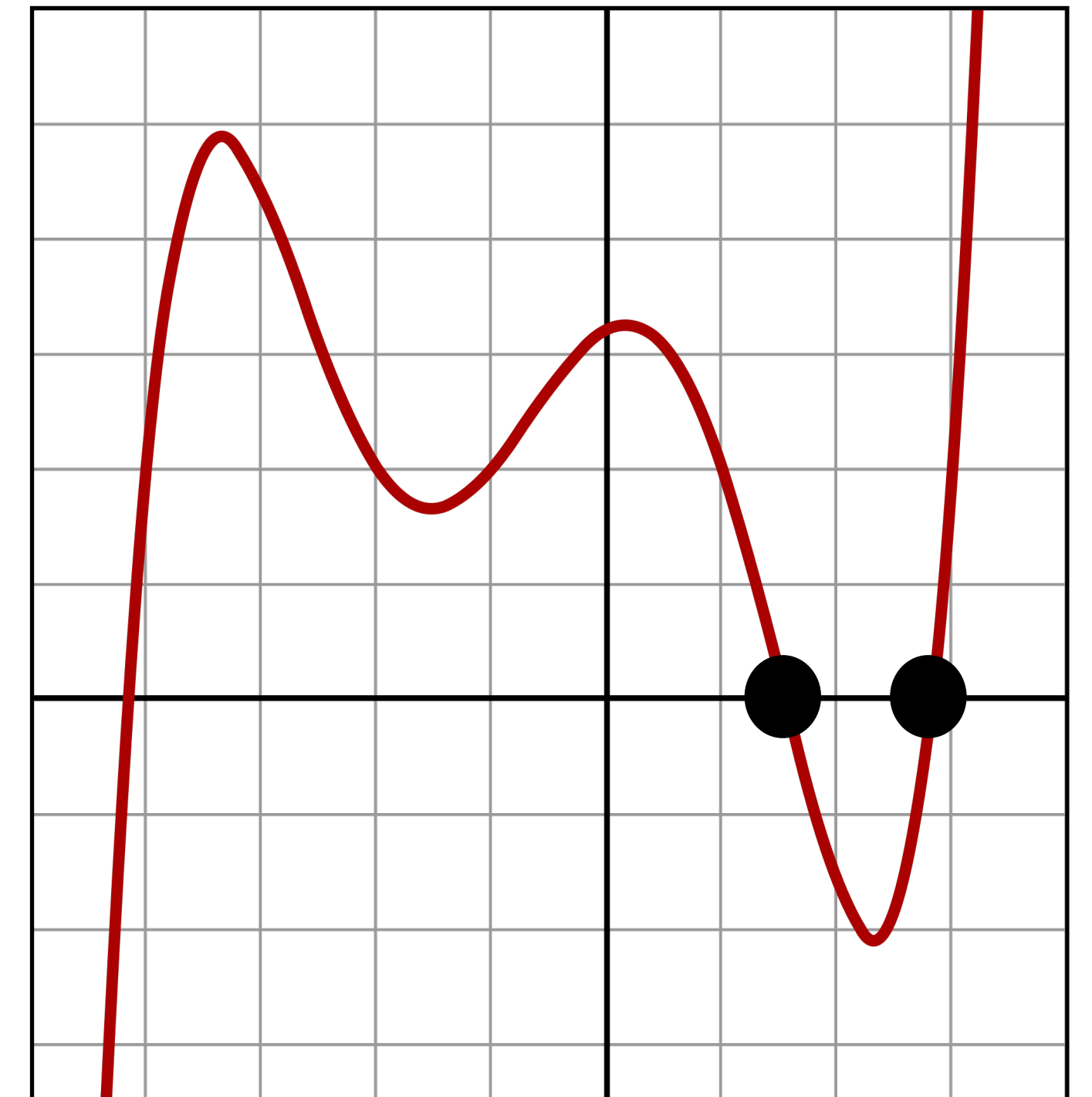
Reminder: Polynomial Roots

A **root** of a polynomial $p(x)$ is a value r such that $p(r) = 0$

(A polynomial may have many roots)

If r is a root of $p(x)$, then it is possible to find a polynomial $q(x)$ such that

$$p(x) = (x - r)q(x)$$



Characteristic Polynomial

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Df. The **characteristic polynomial** of a matrix A is $\det(A - \lambda I)$ viewed as a polynomial in the *variable* λ .

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This is a polynomial with the eigenvalues of A as roots

So we can "solve" for the eigenvalues in the equation

$$\det(A - \lambda I) = 0$$

Example: 2×2 Matrix

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A - \lambda I =$$

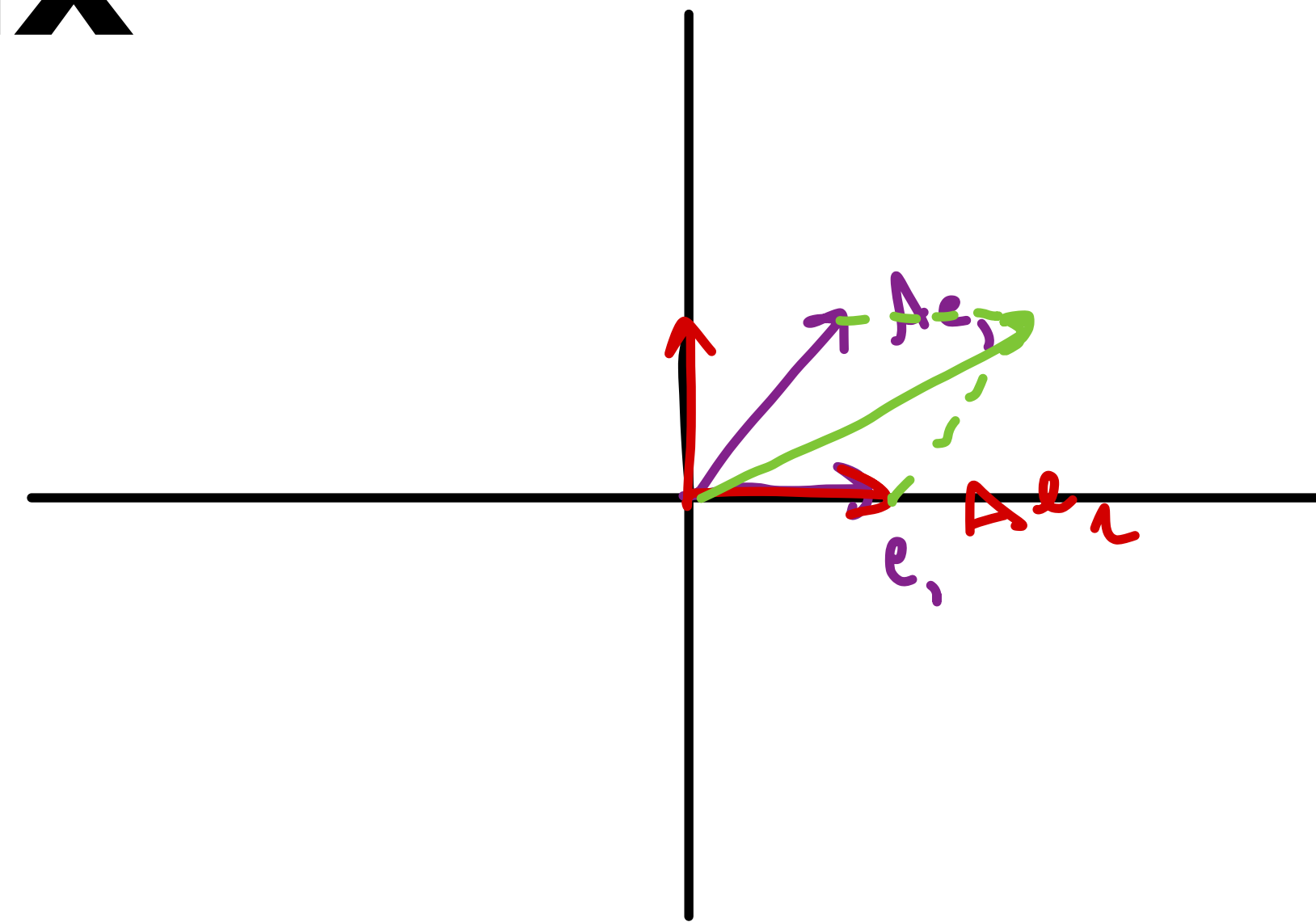
$$\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)(-\lambda) - 1$$

$$= \lambda^2 - \lambda - 1$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}$$

$$\lambda_2 = \frac{1 - \sqrt{5}}{2}$$



$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{1 \pm \sqrt{1 + 4}}{2}$$

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \quad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\overset{k}{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix}$$

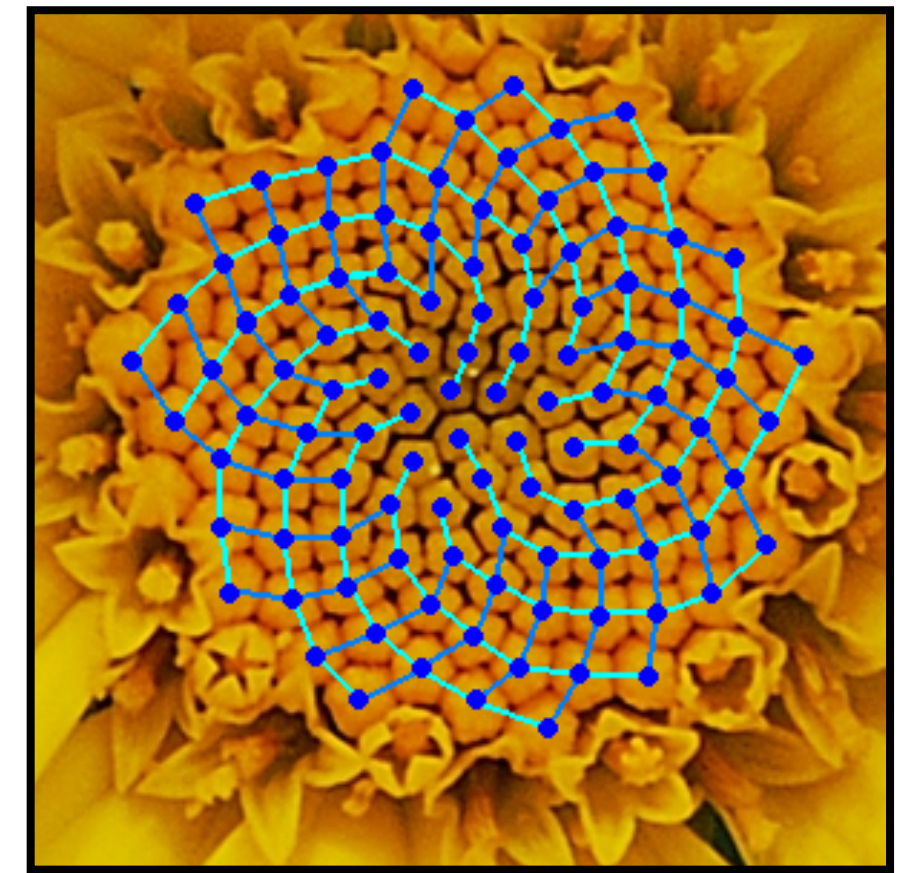
Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

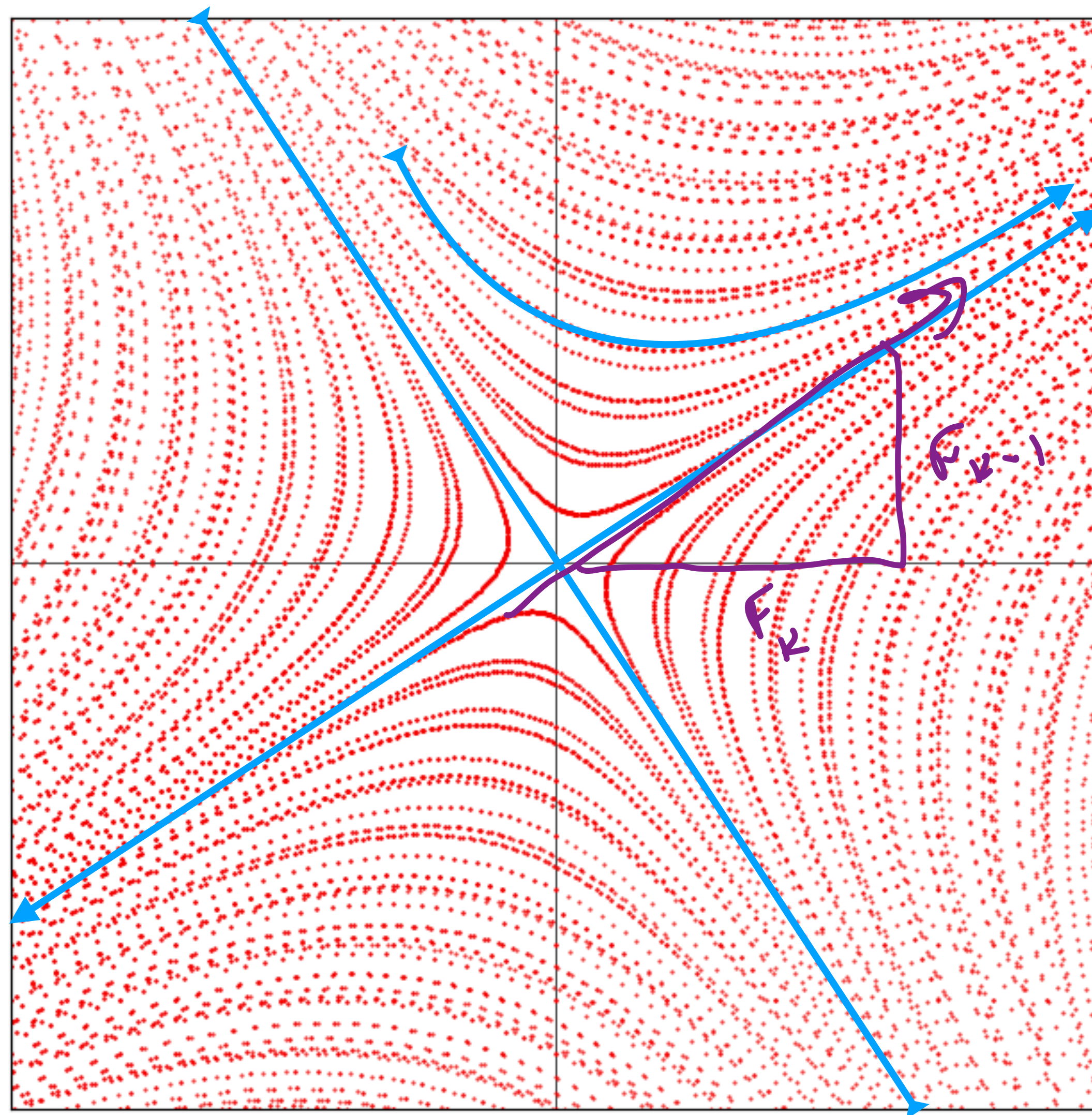
```
define fib(n):  
  curr, next ← 0, 1  
  repeat n times:  
    curr, next ← next, curr + next  
  return curr
```



The Fibonacci numbers are defined in terms of a recurrence relation

They seem to crop-up in nature, engineering, etc.

Recall: The Picture



The eigenvalue of
this matrix is the
golden ratio

Golden Ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

$$\frac{F_{k+1}}{F_k} \rightarrow \varphi \text{ as } k \rightarrow \infty$$

$$F_k \sim c\varphi^k$$

Golden Ratio

$$\varphi = \frac{1 + \sqrt{5}}{2} \qquad \frac{F_{k+1}}{F_k} \rightarrow \varphi \text{ as } k \rightarrow \infty \qquad F_k \sim c\varphi^k$$

This is the largest eigenvalue of $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

The "long term behavior" is the Fibonacci sequence is defined by the golden ratio

Example: Triangular matrix

$$\det(\underbrace{A - \lambda I}_{\text{tri}}) = (1-\lambda)(-\lambda)(1-\lambda)(4-\lambda)$$
$$= \underbrace{\lambda(\lambda-1)^2(\lambda-4)}_{\text{product}}$$

$$\begin{bmatrix} 1 & -3 & 0 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

How To: Finding Eigenvalues

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Q. Find all eigenvalues of the matrix A

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A. Find the roots of the characteristic polynomial of A

An Observation: Multiplicity

$$\lambda^1(\lambda - 1)^2(\lambda - 4)^1 \text{ multiplicities}$$

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Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

geometric multiplicity

Thm. The dimension of the eigenspace of A for the eigenvalue λ is at most the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Example

Let $A \in \mathbb{R}^{5 \times 5}$ with characteristic polynomial

$$(x - 1)^3(x - 3)(x + 5)$$

» What is $\text{rank}(A)$? 0 is not a root $\Rightarrow A$ is inv.
 $\Rightarrow \text{rank}(A) = 5$

» What is the minimum possible rank of $A - I$? ex.